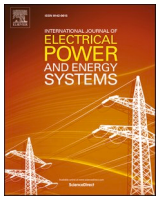
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Review on measurement-based frequency dynamics monitoring and analyzing in renewable energy dominated power systems

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ABSTRACT

The massive integration of clean energy and renewable energy source (RES) is one of the most important trends in the low carbon energy transition toward carbon neutrality. Such an important goal significantly alters the power system dynamic behavior. Consequently, monitoring and analyzing the complex frequency dynamic of RES-dominated power systems play an essential role in ensuring power system security and stability. The measurements, especially those supported by wide area measurement systems (WAMSs), can enable full state awareness and deep informatization of the power system and therefore provide effective solutions for monitoring and analyzing system frequency dynamic behaviors with complex features. This paper focus on the application of measurements in monitoring and modeling analysis technologies for the features of frequency dynamics of RES-dominated power systems. We summarize related research on online monitoring of frequency dynamic behavior features based on the information from measurements. Further, for system frequency security assessment, measurements can improve the traditional methods and promote new data-driven methods for more accuracy and effective frequency dynamics analysis. Finally, future directions and possible solutions for measurement-based frequency dynamics monitoring and analyzing are prospected.

1. Introduction

The growing concern about global warming and the energy crisis has dramatically increased the need for cleaner energy in recent years. In 2020, the world's share of renewables in power generation rose to record levels (11.7%), and Europe's share of renewables reached 23.8%, surpassing nuclear energy and making Europe the first region where renewable energy sources (RES) are the dominant source of power generation [1]. To achieve the goal of "carbon peaking" by 2030 and "carbon neutrality" by 2060, China has set essential steps and main initiatives for the energy and power industry: to construct a new power system of the future, dominated by renewable energy. In China, the share of solar and wind generation is expected to reach more than 25% in 2030, and further increase to 60% by 2060 [2,3]. The energy transition program of Australia allows its power system to securely operate with up to 75% of variable renewable generation by 2025 and nearly 90% by 2035 [4]. The integration of RES into electric grids and their gradual dominance is the present status and future trend of the power system worldwide.

As the power system transitions from a fossil-fuel-dominated system to a clean-energy-dominated system, it will experience the transition from the deterministic power balance of "source following load" to the probabilistic power balance of "source-grid-load-storage" synergy. The system will evolve from a high rotational inertia system dominated by mechanical electromagnetic devices to a low inertia system dominated by power electronics [5]. Accordingly, new security and stability concerns have arisen, which play a vital role in the normal operation of RES-dominated systems. Frequency stability issues are one of these concerns [6]. In recent years, several frequency security related incidents around the world have been attributed to the high penetration of RES [7]:

(1) On September 19, 2015, the "Jinping-Sunan" DC transmission line bipolar locking fault in China resulted in about 4,900 MW of power shortage in the East China Power Grid. The failure caused the system frequency nadir to drop to 49.56 Hz, and the system operate out of the allowed frequency range (below 49.8 Hz) for up to 221 s, indicating the inadequate frequency response capability of the receiving grid [8,9].
(2) The 2016 South Australia blackout is the first known blackout incident in the high renewable penetration system. This state-

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wide major blackout sent around 1.7 million people into darkness and resulted in a financial loss of about 367 million dollars. Right before the incident, wind and photovoltaic provided almost 50% of the total power in South Australia, while conventional synchronous generation (SG) only accounted for 17.6% [10], indicating that the system was in an extremely low-inertia situation. Therefore the integration of RES, which led to a high rate of change of frequency (RoCoF) and frequency dip, was accused of causing this catastrophic blackout [11].

(3) The blackout event of August 9, 2019, in the UK occurred due to several separate and simultaneous events leading to a large amount of sudden power deficiency. This caused a rapid fall in system frequency, which eventually triggered the operation of under frequency load shedding. The loss of nearly 500 MW of embedded distributed generation associated with the transient voltage and frequency disturbance and the reduced energy supply of Hornsea off-shore windfarm due to the operation of protection systems resulted in a large frequency drop outside the normal range occurred [12].

(4) The system separation accident occurred in Continental European Synchronous Area on January 8, 2021, due to cascaded trips of several transmission network elements [13]. As a result of the cascading disconnections, the system was separated into two areas where the South-East area saw an increasing frequency to 50.6 Hz, while the North-West area experienced a decreasing frequency to 49.4 Hz. However, frequency response, automatic interruptible load, and frequency support over HVDC links reacted rapidly after the incident, preventing any additional major damage and outages.

Among all of the events, fluctuations in frequency beyond acceptable ranges led to worsening conditions that eventually resulted in a major blackout, causing significant economic losses and social impacts. When the power system experiences sudden power imbalances due to some external disturbances (such as generator start-up/shutdown, load changes, or failures), system frequency transitions from normal operation to a new steady state or destabilized state, a process called frequency dynamics. Recent frequency-related incidents indicate that the increasing integration of RES without frequency response controlling and monitoring worsens the system frequency dynamics and increases the risk of cascading failure and even frequency instability. However, utilizing the frequency regulation potential of RES generations and demand response is an effective solution for enhancing system frequency stability. Therefore, effective monitoring and analysis of the system frequency dynamic behavior are essential for a RES-dominated power system.

Compared with conventional power systems, the RES-dominated power system has undergone significant transformations in generation, transmission, and demand sides [6], leading to a foreseeable revolution in system frequency dynamics [14]. There have been numerous studies summarizing frequency analysis and control [15–18], Some of which introduce the feasibility and control solutions of frequency response from wind farms [19–21], solar photovoltaic (PV) generations [22], energy storage systems (ESSs) [23], and demand side [24–26]. The integration of converter-based frequency response complexifies system frequency dynamics and raises a series of challenges for frequency stability monitor and analysis of RES-dominated power systems. Therefore, some literature reviews focus on the impact of the increased RES integration on frequency stability issues and potential countermeasures for frequency control [7,27]. Inertia estimation which is the vital topic of system frequency dynamics monitoring due to the low inertia trend in future power systems, has been discussed and comprehensive overviewed in [28,29].

Meanwhile, new tools and techniques have been introduced to deal with frequency problems in RES-dominated power systems, such as deep learning [30]. The basis of these new technologies is the full state awareness and deep informatization of the power system, which relies mainly on measurements to achieve. The supervisory control and data acquisition (SCADA) system is a vital tool for measurement in traditional power systems, but now it is difficult to track fast system change. Therefore, novel measuring devices such as wide area measurement systems (WAMS) and phasor measurement units (PMUs), which have advantages such as high time synchronization, short update cycle, and great measurement accuracy [31,32], have been introduced for real-time monitoring and control of power system states. They have provided solutions for estimation, monitoring, protection, and stabilization of power systems, and this also promotes some research on using measurement information in power system frequency dynamic analysis, as PMUs can detect dynamics that are invisible to traditional meters.

Therefore, this paper aims to provide a comprehensive review of the essential role of measurements in frequency dynamics monitoring and analysis in RES-dominated power systems. Despite the extensive research conducted on this topic, there is a lack of comprehensive summaries. It is worth noting that the machine learning (or called data-driven) applications in [30] are only parts of the measurement-based frequency dynamics analysis proposed in this paper. The applications of measurements also include recording frequency dynamics, post-contingency analysis, system dynamic model identification, frequency security monitoring and prediction. The measurement data mentioned in this paper are primarily obtained from PMU/WAMS with high sampling rates and time synchronization, although there are also studies with data taken from SCADA or even simulation systems.

Overall, this article first summarizes the monitoring of system frequency dynamic behaviors, including inertia, frequency response indicators, spatio-temporal characteristics, and frequency event detection, based on real-time recording measurements and corresponding data analysis. The applications of measurements in system frequency security assessment modeling are then articulated, including how to utilize measurements to overcome the technical limitations of traditional analysis methods and build a new pattern of methods. Furthermore, the paper analyzes the future work needed to explore frequency issues, the possible role of measurements and our suggested technical solutions, providing valuable references for the development of the frequency dynamics monitor and analysis in RES-dominated systems.

The rest of the paper is organized as follows. Section 2 discusses the frequency dynamics in RES-dominated power systems. Section 3 presents the current status of research on frequency behaviors monitoring based on measurements, followed by a discussion of various system frequency security assessment modeling methods and the applications of measurements in Section 4. Finally, Section 5 presents the conclusion and future work.

2. Frequency dynamics in renewable energy dominated power systems

Frequency dynamics is a time series that reflects the power system's ability to restore power equilibrium through system spontaneous response (such as inertia response) and various control actions (such as primary and secondary frequency response) following a significant imbalance between supply and demand [19], as shown in Fig. 1. Several performance indicators from frequency dynamics are used to describe and evaluate the margin of system frequency steady and stability. These indicators include RoCoF at the time of the disturbance, maximum transient frequency deviation (MTFD) $\triangle f_{max}$, frequency nadir f_{min}, and steady-state frequency deviation (SSFD) $\triangle f_{ss}$ [33]. The following factors determine the system frequency dynamics and the magnitude of the related indicators:

(a) The magnitude of active power imbalance in contingency.
(b) The ability of system frequency response, including the total inertia of the system, the amount of active power spinning reserve

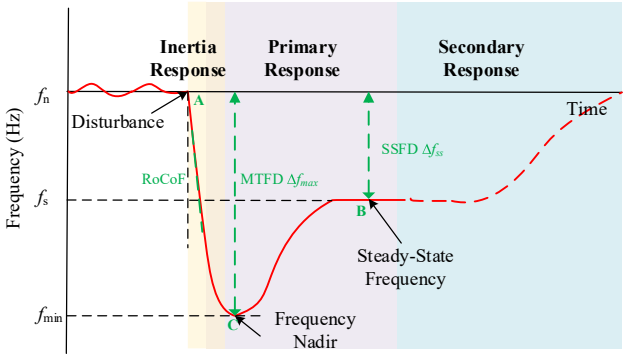


Fig. 1. Illustration of power system frequency dynamics after a disturbance.

available, the speed and reliability of the response of the turbine-governors, and so on.

Since the high penetration of RES, electric power systems worldwide have undergone a significant transformation, predominantly characterized by a proliferation of power electronic converter (PEC) interfaced technologies in generation, transmission, and demand side [6], as illustrated in Fig. 2. Wind and PV generations are generally integrated into the grid by PECs, which leads to converter interfaced generations (CIGs) increasingly replacing conventional SGs in generation side [34]. High voltage direct current (HVDC) transmission has enabled the expansion of RES due to the geographical dispersion and variability of RES, providing flexibility to integrate continuously higher shares of RES while remaining stable [35]. Therefore, with the development and diffusion of HVDC, an AC-DC hybrid transmission grid has become the mainstream mesh, and much more PEC interfaced devices are integrated into the transmission side in RES-dominated power systems [36,37]. As for the demand side, PEC interfaced loads, such as induction motors fed by inverters, electric vehicles, and microgrids [24,38], have progressively gained more shares in modern loads. The penetration of PEC technologies and unpredictable RESs have altered the power system dynamic behavior and arisen new stability concerns [7], which present both challenges and opportunities for system frequency response behaviors.

The challenges posed by the increasing penetration of RES and PEC in power systems are closely related to the two factors that determine frequency dynamics, as mentioned above. Firstly, the range and probability of power deviation have increased significantly. On the one hand, large-capacity HVDC transmission leads to a higher probability of large power shortage events following a single failure in the receiving-end power grid, as the HVDC blocking faults can produce a huge transient energy impact. On the other hand, it is more difficult to balance the demand/supply chain as RESs, particularly wind and solar energy, are intermittent and stochastic, which increases the stochastic variation of active power in generation sides [39–41]. Secondly, the ability of system frequency response has been degraded [42]. The power system's inertia has persistently decreased as SGs have gradually been replaced by CIGs, which have little or no inertial response [15]. Meanwhile, an increase in the penetration of RES and HVDC, leading to the scarcity of conventional generators, results in a decline in power reserves for frequency control, further decreasing the system's ability to respond adequately to frequency events. Hence, the challenges faced by frequency dynamics in RES-dominated power systems are that the attacks (the range and probability of power deviation) are becoming stronger while the defense (the ability of frequency response) is becoming weaker. The combined effect increases the risk of large frequency deviations and even frequency instability for severe disturbances [43].

PECs have a variety of advantages in independently controlling the active and reactive power being transformed with the grid, and their overall performance is dominated by the control systems and strategies of the converters. Therefore, the entire power system chain, including generation, transmission [44–46], demand [24,47] and storage side [48,49] has significant potential in providing regulation power and frequency control support [17,50]. Many grid operators around the world have issued grid codes requiring RESs and PECs to participate in frequency control as needed as a system service [51]. However, the inherent mechanisms of PECs providing frequency response differ significantly from those of SGs [14]: for SGs, the primary power for frequency regulation is fully controllable and the frequency regulation strategies and parameters are similar, but the stochastic nature of RES means that the primary power of RES for frequency regulation is uncontrollable, and the strategies of converters can be set adaptively to their own nature.

Heterogeneous and multi-scale frequency response in systems with PEC will bring much more complex phenomena to system frequency dynamics, such as second frequency drops due to the need to terminate frequency support from RES when the limited energy stored in their storage elements runs out [52]. New features of frequency dynamics

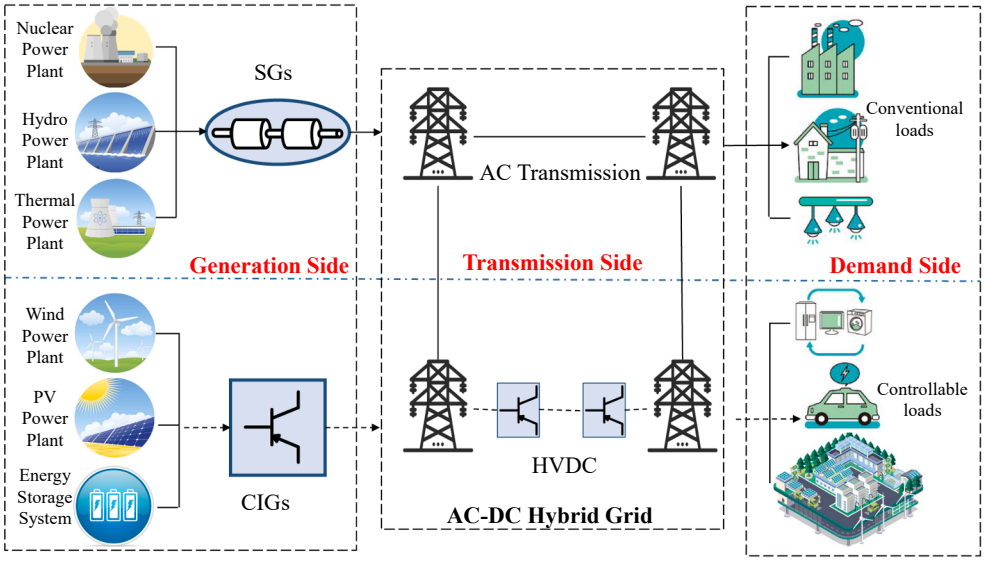


Fig. 2. Illustration of the penetration of PEC in RES dominated power system.

have arisen that need to be appropriately analyzed, characterized, and monitored. The traditional analysis and modeling methods are not sufficient to cope with frequency problems in RES dominated systems, which are high-dimensional time-varying nonlinear power systems with multi-source interactions [30]. Revealing the intrinsic dynamics of frequency problems from a data perspective has progressively become the trend in system frequency analysis.

3. Measurement-based frequency dynamic behaviors monitoring

Frequency dynamic behaviors monitoring is critical for assessing frequency security assessment [53], coordinating frequency control [54], and configuring frequency regulation sources [55], especially in power grids integrated with high shares of RES. The monitoring involves real-time recording and analysis of system frequency during normal operation or accidental disturbance. Real-time phasor data from PMUs provide valuable information on power system dynamics that is appropriate for frequency dynamics monitoring.

Accordingly, measurement-based frequency dynamics monitoring involves extracting effective information that reflects the system frequency features from abundant PMU measurement data in a wide area system (some studies also refer to SCADA measurements or simulation test data). The frequency dynamic behaviors monitoring can be divided into several categories based on frequency features: inertia estimation, frequency response indicators estimation, spatio-temporal characteristics monitoring, and frequency event detection. These frequency behavior features describe the entire process of frequency deviation occurrence, development, and propagation in the power grid after a disturbance occurs. The monitoring of these frequency behavior features is not absolutely independent, but interrelated, as shown in Fig. 3. They are discussed in detail from Section 3.1 to Section 3.4.

3.1. Inertia estimation

The inertia of a power system represents the energy stored in the power system that can be released to provide frequency support during disturbances. In terms of the discussion in Section 2, the penetration of RESs makes the system inertia variable and changes the distribution of inertia across the grid, leading to the formation of low inertia areas [56,57]. The low and fluctuating values of inertia could jeopardize frequency security, thus making continuous monitoring of inertia across the system essential for the power system's reliable operation. Various devices, including synchronous machines, induction motors, and RESs, contribute to the system inertia [58,59], making it unviable to obtain accurate system inertia simply by aggregation [60]. However, the extensive deployment of PMUs facilitates online evaluation of inertias by providing synchronized measurements, e.g., power and frequency, at different locations in real-time, also named measurement-based inertia estimation.

The existing works on measurement-based inertia estimation can be classified according to two dimensions: operating conditions and methods used. In terms of operating conditions, the inertia can be evaluated from the measurements during disturbances or normal operation (ambient signals). Besides, approaches that include the injection of a probe signal that excites the dynamics of the system are discussed [61,62,63]. As disturbances are rare, disturbance-based methods could not monitor the inertia consecutively. This problem could be alleviated by the latter two categories.

The mainstream methods used in inertia estimation are shown in Fig. 4 and discussed as follows.

(A) Direct methods

When disturbances occur, the mechanical powers remain virtually unchanged, so a straightforward way to calculate the inertia is as shown in Eq. (1):

$$M = -\Delta P / \dot{f}\Big|_{t=t_c} \tag{1}$$

where M represents system intertia constant, $\triangle P$ is the imbalance power, $\dot{f}$ is the differentiation of dynamic frequency, and t_c denotes the time of disturbance.

For direct methods, difficulties lie in detecting the disturbance, accurately calculating the RoCoF, and estimating the power imbalance. The occurrence of a disturbance is currently identified according to rules of thumb, e.g., when RoCoF exceeds a threshold [64,65] or when the estimated inertia value stops fluctuating [66]. Filtering [60,67], polynomial curve fitting [64,65], and simplified linear fitting techniques [68] are adopted to approximate the frequency to mitigate measurement errors. The RoCoF is then calculated by differentiating the approximated frequency. Energy-base methods are proposed as a substitution to avoid inaccuracy in RoCoF estimation, where inertia is obtained by dividing the energy provided by the inertial response by the frequency deviation

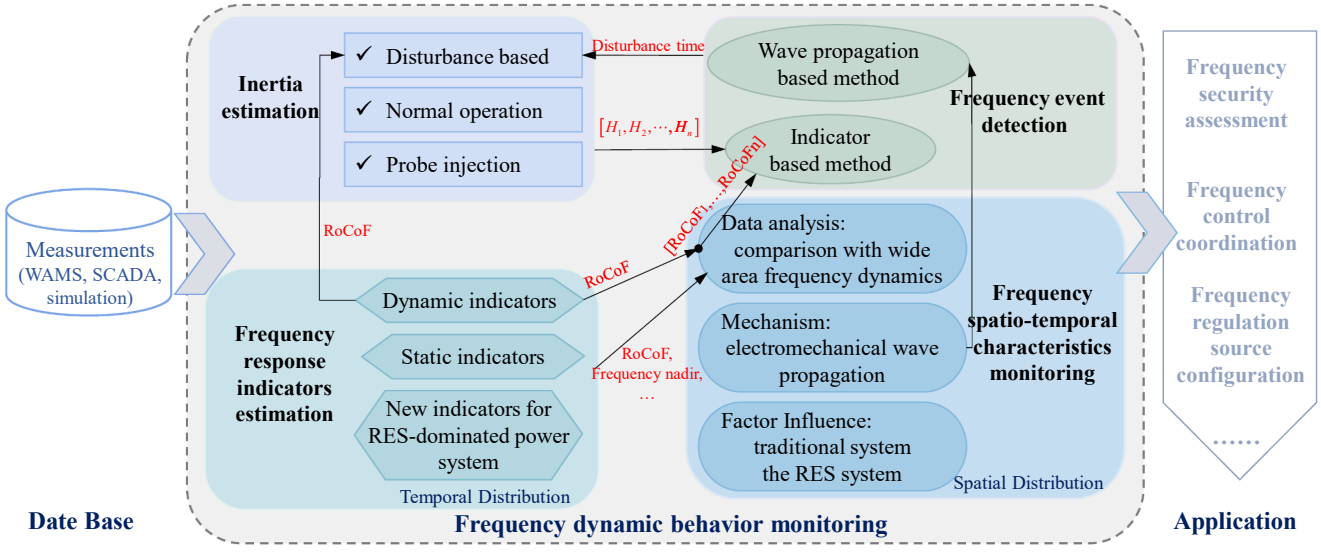


Fig. 3. The of measurement-based frequency dynamic behaviors monitoring.

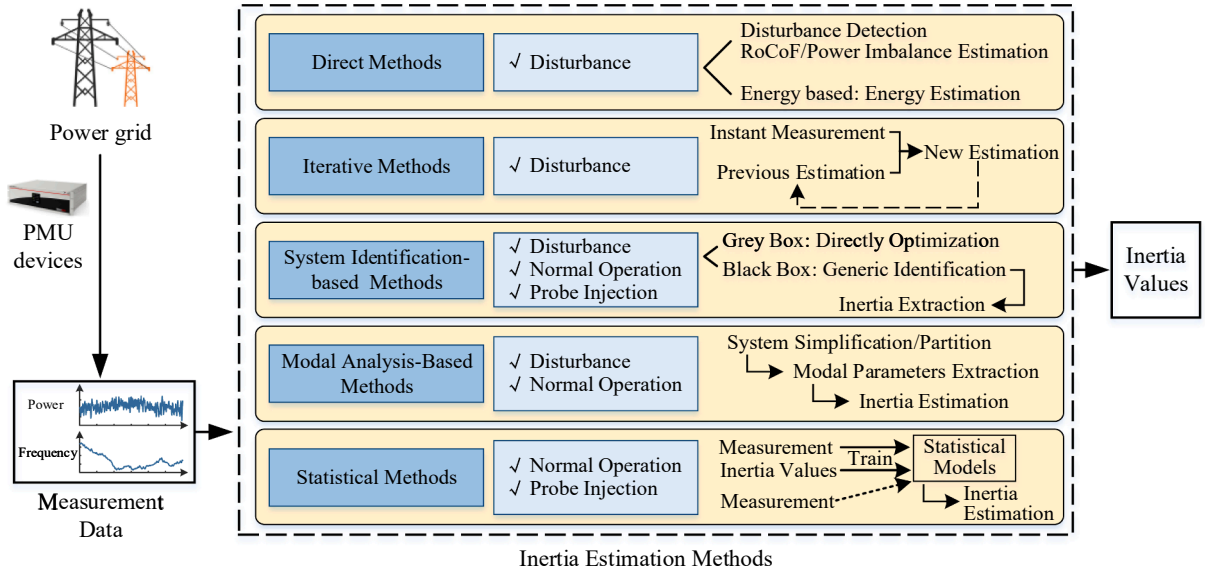


Fig. 4. Diagram of measurement-based inertia estimation.

after a disturbance [69,70]. In most literature, the power imbalance is assumed to be known. To estimate the power imbalance accurately, [71] considers the load response caused by voltage and frequency variations, while [65] employs curve fitting techniques.

Nevertheless, the conspicuous drawback of many direct methods is they are incapable of discerning non-synchronous inertia, as revealed by [72] and [65].

(B) *Iterative methods*

The essence of iterative methods is to adjust the estimation results continuously and iteratively until reaching an equilibrium determined by the swing equation. Various filter techniques have been proposed [73,74]. A classic iterative tool is the Kalman filter, which treats inertia as an iteratively updating state variable. Methods based on the extended Kalman filter [75] and unscented Kalman filter [76] are also discussed. The iterative nature of these methods allows for inertia estimation updates at each time step during disturbances, resolving the latency issue of direct methods. Besides, their competence in tracking non-synchronous inertia is gradually manifesting its significance.

(C) *System identification-based methods*

System identification, which includes gray box identification and black box identification, is a powerful tool for modeling the dynamic characteristics of complex systems through measurements. Gray box models consist of unknown parameters with explicit physical meanings, and their structures follow those of validated physical models. [62] presents a moving-horizon-estimation-based algorithm that optimizes the inertia to fit the measurements. [77] utilizes gray box models to characterize the non-synchronous inertia and considers nonlinearity by integrating nonlinear Hammerstein-Wiener models. In black-box-model-based methods, the dynamics between power and frequency measurements are firstly fitted by generic models, e.g., ARX [78], ARMAX [79,80], or subspace models [61,81], without considering specific physical mechanisms. Inertia is then extracted from these models, e.g., by analyzing the impulse [79], step response [78,81], or frequency response fitting techniques [61]. Recently, black box identification has been also adopted to model non-synchronous devices, e.g., doubly-fed induction generators (DFIGs), or voltage source converters [82,83].

(D) *Modal analysis-based methods*

The equivalent inertia of complicated power systems can be determined by accurately identifying model parameters, as the oscillation modes are related to the equivalent inertia when the electromechanical dynamics are approximated by equivalent swing equations. Model analysis-based methods simplify the estimation of power imbalance and toCoF, which are cumbersome in the direct method. Using measurements during a disturbance, [84] puts forward an approach that simplifies the system to a two-generator equivalent system and extracts oscillation parameters. Furthermore, the Fourier transform is introduced to enable the inclusion of multiple modes [85]. Methods that use ambient measurements to estimate the effective inertia of each area concurrently by analyzing the inter-area oscillation modes are developed in [86] and [87].

(E) *Statistical methods*

Some works propose statistical methods to estimate inertia, where inertia is calculated based on analyzing batches of historical data rather than on physical laws. Statistical tools comprising Markov Gaussian models [88], stochastic processes [57], and neural networks [63,89] are utilized. Statistical methods require numerous training data with inertia values known in advance, which limits their practical applications. Furthermore, as the swing equation is not regarded, the issue of interpretability still exists.

The development trend in measurement-based inertia estimation methods could be summarized as follows. New approaches are constantly emerging, catering to different situations. The need to estimate the inertia online, especially during normal operations, is gaining urgency. The impact of RESs is becoming not negligible, and the attention on the estimation of non-synchronous inertias is on the increase. The influence of measurement noises is progressively taken into account [62,63,69,73,76,78,80,82], and the need for noise-robust algorithms is rising. The inertia is considered in more detail, with its time [57] and space [90,91] distribution noticed.

3.2. Frequency response indicators estimation

The relationship between frequency response trajectory and the given requirement during a disturbance reflects the frequency security of the system [92]. Therefore, extracting and analyzing frequency

trajectory information (i.e., frequency response estimation) is an important reference for system frequency safety assessment. The key to frequency response estimation is to correctly identify the indicators that reflect frequency behavior performance. According to the classification of frequency security, a complete indicators system includes two aspects: 1) Transient (dynamic) indicators, which are concentrated on the instantaneous frequency changes that contain the starting time of a disturbance, RoCoF, frequency nadir, and the time to reach the nadir; 2) Steady-state (static) indicators, which include steady-state frequency and the time to reach it [33]. Different indicators represent various periods or states after incidents and are impacted by various system characteristics, as summarized in Table 1. Single or multiple indicators above are usually estimated to represent the frequency dynamic trajectory information, and then calculate their distance from the security requirement limits to access the system frequency security [53,93]. The relation between the indicators and the influential factors can also guide gird operators to adjust the frequency control and reserve to ensure the safe operation of the system. Also, the assessment results are gradually being applied to RES integration planning as limit constraints in modern power systems [55]. However, these indicators that are traditionally used in SG-dominated systems are insufficient at reflecting the support from the RES and PEC. Some new indicators have to be introduced to describe the complex frequency dynamic in RES-dominated systems, such as the maximum deviation and starting time of the secondary frequency drop, which occur in systems of wind plants providing frequency response [14].

The accuracy of frequency response indicators estimation from PMU measurements is reduced due to ineluctable noises, bad data, and spikes from the real frequency measurements [94]. Some researchers have proposed more precise estimation algorithms: [95] proposed a response starting point estimation method with a second derivative of frequency data, and RoCoF estimation with flexible window size for low inertia power grids in [96]. The frequency response estimation problems will be especially difficult in RES integrated systems with complexity frequency dynamics, therefore, new estimation algorithms should take full account of the requirements of the features of the RES-dominated system.

3.3. Frequency spatio-temporal characteristics monitoring

The frequency dynamics of different buses in the system actually vary during a disturbance. At the moment a disturbance occurs, the power shortage will be distributed among SGs based on the electrical distance between SGs nodes and the disturbance location. This distribution will cause deviations between the speeds of different SGs, resulting in differences in the frequency response of each node. And then the changes in the voltage phase angle deviation will affect the power flow of the system, which redistributes the power shortage until the frequency of each node reaches synchronization again. The propagation of disturbances in the power grid makes it inevitable that the spatial distribution of frequency dynamics will occur. In the past, the frequency difference between generation buses was negligible, and the spatial characteristics of frequency response dynamics were usually ignored, as the frequency changes slowly in the contingency due to the sufficient rotational inertia in conventional power systems, and in which the distribution of the amount of inertia and frequency response reserve is even as the SGs are the main frequency regulation resource. The Center of Inertia (COI) frequency was extensively applied to represent the average system frequency [97].

Table 1
Frequency response indicators estimation.

Categories	Indicators	Influential factors	Representing state
Transient (dynamic)	Starting time RoCoF	Disturbance occur inertia	Initial state
	Frequency nadir Frequency nadir time	Inertia & frequency regulation	“worst” state
Steady-state (static)	steady-state frequency steady-state frequency time	Damping & energy reserve	Initial steady state
Secondary frequency drop (SFD)	Maximum deviation of SFD Starting time of SFD	PEC frequency response performance	Additional state due to PEC integrated

However, the increasing penetration of RESs and PEC has significantly impacted the spatial characteristics of system frequency. Firstly, the degradation of the system total inertia and frequency regulation ability will cause a sharp change in frequency when the system suffers a disturbance, resulting in significant differences in the RoCoF of each bus during the initial period of perturbation [98]. And then PEC devices with frequency response control centralized or distributed integrate into the system, leading to heterogeneous, non-uniform frequency regulation resources and inertia distribution [99]. Furthermore, cross-area HVDC transmission lines divide the system into sending-end and receiving-end power grids, which will experience the exact opposite power shortage while HVDC is tripped [42]. As the result, the frequency dynamics of the two areas will be different. These factors illustrate that the spatial distribution of frequency dynamics is no longer negligible, and the assumption of COI frequency can lead to significant errors in frequency analysis and control in RES-dominated power systems.

A significant amount of research has been conducted to observe and quantify the spatio-temporal characteristics of frequency dynamics in power systems using dynamic simulation or real measured data. The main scheme of this research is shown in Fig. 5. The comparison of wide area system frequency behavior during disturbance reveals that different locations would have different frequency responses during and after a disturbance [100]. Global positioning system(GPS) time-synchronized PMUs can continuously monitor frequency dynamics at different locations, providing valuable data for monitoring frequency spatio-temporal characteristics. Frequency monitoring networks (FNETs) based on PMUs have been implemented to achieve synchronized observation of the entire U.S. power system [101]. Analysis of data collected from the FNET shows that the frequency dynamic indexes (such as RoCoF, frequency nadir, and the time at frequency nadir) exhibit significant variability in different locations [102,103]. The process of frequency fluctuations changing from local independent oscillations to overall homogeneous behavior has been observed with an open database of grid frequency measurements across 17 locations on three continents [104]. Electromechanical wave propagation phenomena are considered to be the internal mechanism of frequency dynamic spatial distribution [100,102]. The frequency wave will propagate at different speeds depending on the location and direction of observation, and the anisotropy of the frequency propagation speed can be used to realize the disturbance location [105]. In addition to the time delay of frequency deviation [106], some statistical model methods have been investigated to describe the propagation model of frequency dynamics [107].

Various factors impact the frequency spatial distribution, including the disturbance location, network topology, line impedance, power flow, inertia distribution, and frequency regulation distribution [98]. With the high penetration of RES, the dynamic coupling among areas and the inertia heterogeneity will lead to substantial differences in the frequency dynamics of individual areas [99,108]. The RES response characteristic, frequency and voltage interactions and other factors also affected frequency dynamics distribution in space. Monitoring wide area system frequency with spatio-temporal characteristics helps transmission system operators understand system frequency security and stability and better allocate the speed and quantity of limited frequency response services.

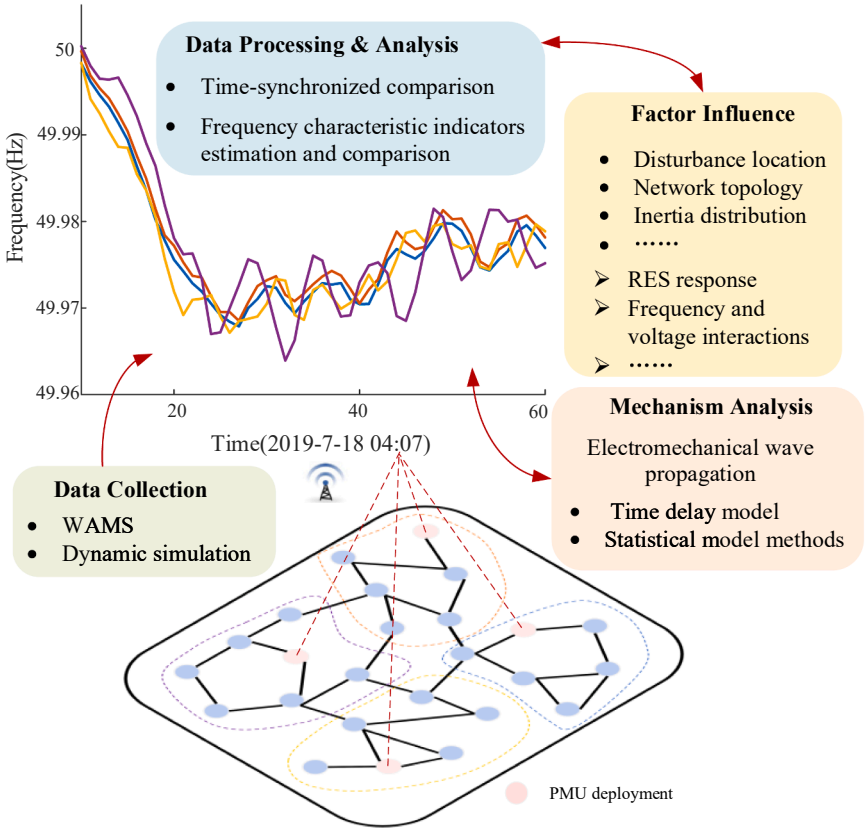


Fig. 5. The main scheme of frequency spatio-temporal characteristics monitoring based on measurements.

3.4. *Frequency event detection*

Event detection, which aims at identifying the inception time, location, and severity of an event, is the key to active preventive protection controls to prevent cascading failure. A frequency event is a disturbance caused by power mismatches between generation and demand resulting from generator/transmission line tripping or load disconnection [109]. The impacts of such events have become increasingly critical with the increased penetration of renewables and PEC, as evidenced by major disturbances in power systems across the globe (discussed in *Section I*). WAMSs have been considered an effective tool for identifying risks in some countries as they provide high-resolution and time-synchronized PMU data that can monitor sharp dynamic information throughout the network [110,111]. Naturally, frequency measurements in wide area are used to localize power disturbances, which can be summarized as indicator-based methods and wave propagation-based methods. The indicator-based methods determine the disturbance location by comparing specific indicators estimated in various locations, such as selecting the minimum event starting time (EST) from all frequency disturbance recorders [112], identifying the data stream with the highest strength using self-defined energy deviation [113] or wavelet energy [114,115]. The indicator RoCoF is also used to assist in estimating power imbalance amount as (2) [116], where P_{Event}, P_{gen}, and P_{load} are the imbalance power, total generation power, and total load power, respectively; f_N is the nominal frequency; H_i and S_i are inertia constant and capacity of the $\#i$ generator; $\sum H_i S_i$ represents the total inertia from the N_{Gen} generations of the system, and $\dot{f}$ is system dynamic frequency differentiation, also known as RoCoF. These simple algorithms can avoid significant delay from advanced algorithms and thus are widely used in practical systems. The rapidity of frequency event localization is of paramount importance, as algorithms with minimal delay can secure time for subsequent control actions, thereby effectively mitigating the escalation of events. Therefore, even though there might be more complex and accurate methods for achieving localization, simpler computational approaches often prove to be more effective in practical systems. However, the indicator-based methods heavily depend on the results of the inertia estimation and frequency response estimation, which have been discussed as another challenge in frequency dynamics monitoring in *Section 3.1 and 3.2*.

$$P_{Event} = P_{gen} - P_{load} = \frac{\sum_{i \in N_{Gen}} H_i S_i}{f_N} \dot{f} | t = t_c \tag{2}$$

According to the formation mechanism of frequency space distribution, electromechanical wave propagation theory is applied to disturbance localization, named wave propagation-based methods. The major steps of these methods include: detecting the disturbance arrival time t_i at different locations through WAMS measurements, constructing a set of equations about the wave propagation with geographic coordinates (x_i, y_i) of the hypocenter as (3) [117], and solving the equations by Newton's method or least squares method to obtain actual beginning time t_h and geographic location (x_e, y_e) of the disturbance [118,119]. The implementation of the method is based on the assumption that the disturbance wave propagation speed V remains constant throughout the system, which does not hold true in practical scenarios and will lead to localization errors. To address this issue, [105,120] have considered the anisotropy of wave propagation speed to improve the mathematical model for disturbance localization. Additionally, [121] calculates the disturbance propagation speed distribution from the wave arrival time to determine the event location, while [122] estimates the propagation path based on the time difference of event arrival. Wave propagation-based method can directly locate the geographic location of the disturbance source without additional information about the network topology. However, this approach can also result in localization results that are not on the specific electrical bus, which is not conducive to

subsequent control.

$$(x_i - x_e)^2 + (y_i - y_e)^2 = V^2(t_i - t_h)^2 \tag{3}$$

In power disturbance localization with frequency measurements, monitoring every frequency requires redundancy in practical scenarios. To overcome this limitation, [123] proposes a method that estimates the disturbance size and location based on active power measurements and the system impedance matrix with only a limited number of PMUs, and [124] identifies the size and location of generation loss events using voltage and current phasor measurements from sparse PMUs. However, it is important to note that these methods require the system topology, such as the impedance matrix, as they are only a limited number of PMUs available.The summary about methods of power disturbance identification and localization can be seen in Table 2.

4. Measurement-based system frequency security assessment modeling

4.1. The methods of system frequency security assessment modeling

The methods of system frequency security assessment modeling aim to predict the frequency curve or frequency characteristics of a power system under expected faults to determine its current frequency security margin. Model-driven analysis methods are first introduced to achieve this prediction work, and they are generally divided into two categories: time-domain simulation (TDS) methods and equivalent frequency model (EFM) methods [30]. The framework of these model-driven methods is shown in Fig. 6.

The TDS methods, which are based on numerical analysis, can accurately simulate the real power system by considering the complex control characteristics of various types of dynamic devices, network structures, and the coupling effects of all state variables, such as frequency, power angle, and voltage. The main approach is to establish detailed models of each component and network topology of the system, form a set of differential–algebraic equations, and then solve the system state quantities step by step after a specific disturbance, using the original steady state as the initial value to obtain the frequency time-domain changes. TDS methods are the most accurate methods for frequency dynamic analysis [55] and form the basis of some commercial simulation software such as PSASP, PSS/E, and BPA. However, TDS methods have some limitations, such as requiring a large amount of computation and being challenging to model, which make them unsuitable for online application in actual power systems. They are better suited for offline analysis. The drawbacks are more striking with the high penetration of RESs and PECs, as their control characteristics are more dependent on (complex) fast-response power electronic devices. Even electromechanical-electromagnetic hybrid simulations are needed to obtain the frequency dynamics of RES-dominated power systems [6,125,126].

Therefore, EFM methods are proposed as fast methods, which construct a simplified power system frequency dynamic model by ignoring some weakly frequency-related quantities, such as voltage dynamics, network topology, and nonlinear characteristics of the governor. The average system frequency (ASF) [127] model and system frequency response (SFR) [128] model are typical and widely used models of EFM methods. The SFR model describes the average response of multiple generators with a simplified and aggregated single-machine model [129] and can provide an analytic time-domain expression as (4) for system frequency dynamics. In (4), $\triangle P$ is the power imbalance, the governor speed regulation R, and the damping factor D are parameters of the SFR model, and the natural oscillation frequency ω_n, the damped frequency ω_r, the damping ratio ζ, and the coefficients ϕ are calculated from SFR model parameters, the details are shown in [128]. Based on this feature, the SFR model has been applied to a wide variety of studies related to system frequency dynamics, including planning [130], operation [131,132], and control [47,133]. However, the SFR model is significantly different from the actual model, leading to a lack of accuracy in the calculation of frequency dynamic analysis. As the new need and features of modern system frequency dynamics appear, several works have proposed extending the SFR model: considering the effect of the thermal state of the generator [134] or the non-linear characteristics of the governor [135,136], taking into account the heterogeneity of multi-machine systems [137,138], extending to quantitatively analyze the spatial characteristics of frequency dynamics [139–141], and most important, modeling the inertia response and frequency response of RESs and PECs [142–145]. Thus, the EFM methods based on mathematical analysis have the advantages of clear mechanism, little computational costs, and easy expansion. However, there are still many deficiencies in practical applications, especially for bulk power systems, such as low solution accuracy and difficulty in obtaining equivalent parameters.

$$\Delta f(t) = \frac{R\Delta P}{DR+1}[1 + \alpha e^{-\zeta\omega_n t}\sin(\omega_r t + \varphi)] \tag{4}$$

4.2. The application of measurements in system frequency security assessment modeling

With the development of modern power systems, both model-driven methods have inherent defects that are insufficient to meet the requirements of both accuracy and speed for online system frequency dynamic analysis. Fortunately, the advent of wide area measurement information provides situational awareness of the power system, allowing for new ways to overcome the technical bottlenecks of system frequency security assessment modeling. The applications of measurements can be classified into two main categories: improve model-driven methods and arise data-driven methods.

(A) *Improve model-driven methods*

The large amounts of real-time operation data available from WAMS have been extensively used in power system security analysis and operation control. This includes applications such as state estimation [146,147], parameter identification [148–150], and dynamic equivalence [151]. These applications can be directly or indirectly implemented to improve the traditional model-driven system frequency assessment modeling methods.

Firstly, in TDS methods, accurate and reliable system models are crucial to ensure the effectiveness of simulation-driven dynamic analysis [152,153]. Thus, techniques that use measurements to online validate or calibrate dynamic simulation models and their parameters have been

Table 2
The methods of power disturbance identification and localization.

Type of data	Methods	Requirements for PMUs	Advantage	Disadvantage
Frequency measurements	Indicators-based method	PMU measurements redundancy	Less calculation	Relying on the system model or specific parameters
	Wave propagation-based method		Direct to the geographic location	Not located in the specific bus
Active power measurements or voltage measurements	~	Only a limited number	Robustness against partial PMU unavailable	Relying on grid topology

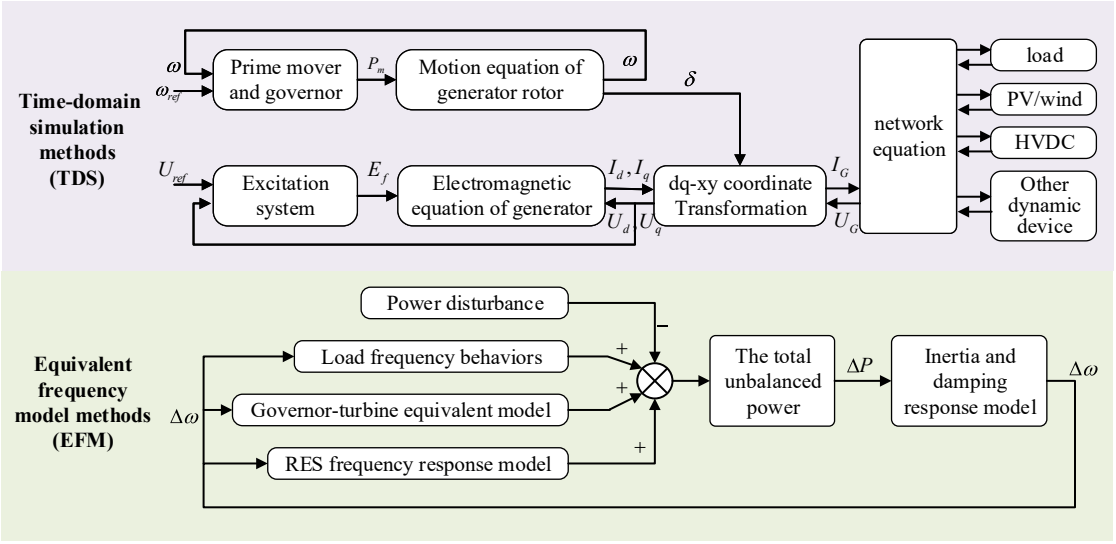


Fig. 6. The framework of model-driven system frequency assessment modeling methods.

required in the smart grid [149,154]. Model validation typically adapts hybrid dynamic simulation (HDS) techniques, which inject PMU measurements into the traditional dynamic simulation to compare the model's response to actual measurement signals [155]. Once model discrepancies are observed, model parameter identification is used to calibrate the model of interest. In the TDS methods, one of the most vital elements is the power plant model, and many studies focus on parameter identification of generating units. Literature [156–158] estimated the physical parameters of synchronous machines from online measurements. However, besides synchronous generators, a complete dynamic model of a power plant also includes various types of controllers, such as Turbine Governor (TG), Automatic Voltage Regulator (AVR), and Power System Stabilizer (PSS). A relatively full set of dynamic state variables and parameter estimation is attempted in [159–161]. As the number of parameters to be simultaneously identified increases, there may be problems such as identification algorithm non-convergence or multiple solutions of model parameters. Therefore, the selection of key parameters or problematic parameters based on trajectory sensitivity analysis is the previous step for parameter identification in model calibration [162–164]. As mentioned above, the framework of the model validation and calibration based on measurements is summarized in Fig. 7. In fact, the identification of partial parameters cannot guarantee the accuracy of the simulation models. Therefore, some researchers have proposed selecting corresponding generator components (PSS, AVR, or TG) with precise models to identify parameters based on different stability problems (such as power angle, transient, voltage, or frequency problems). As we know, TG is the main controller that affects the frequency behavior of the generators. So the model and parameter of TG are vital for frequency dynamic analysis based on TDS methods. [160] carried out an UKF-based estimation to identify the model of the generator's swing dynamics and TG model with primary and secondary frequency control. [165] used a *meta*-heuristic algorithm to validate the IEEE standard models of the TG and exciter separately. And [166] took into account the effect of nonlinearities of the governor and estimated the governor droop and the dead-band with UKF. [167] proposed a closed-loop model for simultaneous parameter identification of SG and TG, considering that the output of the TG (mechanical power) cannot be measured.

Besides conventional generators, dynamic models of power systems, particularly those dominated by REs, include loads, transmission

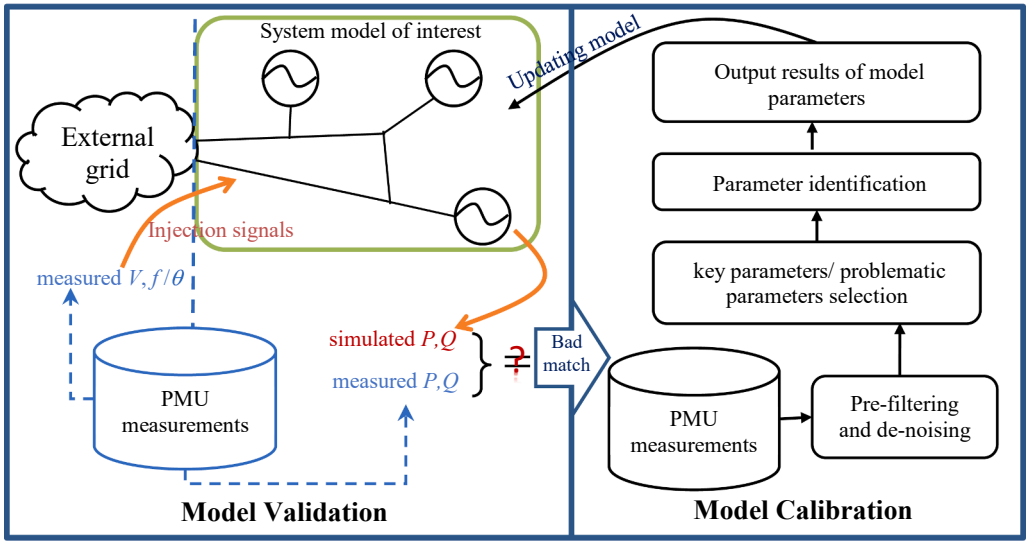


Fig. 7. Dynamic model validation and calibration based on PMU measurements.

systems (both AC and DC), CIGs and their controls. In recent years, plenty of work has been carried out to develop accurate load models for tracking the time-varying and distributed properties of power loads online [148,168–170]. This is a major research field in power systems, but as it is not strongly related to our topic, so we will discuss it here. For more details, please refer to [171]. Regarding CIGs, taking wind power as an example, some studies have focused on identifying electrical parameters [172,173], and estimating converter control parameters [174] of an accurate model of a single wind turbine, while others have concentrated on dynamic equivalent modeling and parameter identification for multi-machine wind power plant based on measurements [175–178]. In general, CIGs have more complex controls and a larger set of parameters, making them more challenging to identify models [179]. Therefore, there is little research on the parameter identification of additional control for wind turbine participation in system frequency response.

Secondly, for EMF methods, measurements can be used to improve the adaptability of the equivalent models. [180,181] develop a generic SFR model structure that can represent power systems with thermal, hydro, and renewable generation. Also, the parameter estimation strategy is proposed based on dynamic and steady-state data to guarantee the effectiveness of the generic model. The method of directly identifying the system frequency equivalent model based on measurements bypasses the challenges of obtaining individual component parameters and constructing models. This approach is better suited for power systems with multiple frequency response resources. [134] have updated the extending SFR model (which considers boiler thermal states) by calculating equivalent static model parameters and thermal states based on wide area measurements. Accordingly, we can see that the measurement-based parameter estimation can overcome the drawbacks of the EMF methods, which is that the parameters are difficult to obtain.

(B) *Arise data-driven methods*

In addition to improving traditional frequency dynamic analysis methods, wide-area measurements can also be directly mined for information to develop "model-free" intelligent methods based on data-driven technology. These intelligent methods are capable of extracting features and classifying high-dimensional heterogeneous multi-source information, making them effective tools for fast online calculation of system frequency dynamics [30]. The process involves feature extraction and offline learning on historical measurements or simulation data, followed by inputting the operation data and expected faults into the trained model to achieve fast online calculation of frequency dynamic performance indicators. Machine learning technologies, such as Back propagation (BP) [182,183], regression tree (RT) [184], support vector machine (SVM) [185,186], and extreme learning machine (ELM) [187], have been widely adopted to frequency dynamic predication and frequency security assessment. However, these shallow learning algorithms have limited feature extraction ability under complex conditions, resulting in the loss of an amount of valuable information and a reduction in frequency analysis accuracy. Deeping learning(DL) can automatically extract features of data via multiple hidden layer structures and thus solve high-dimensional nonlinear complex problems of power systems. [188,189] applied the deep belief network (DBN) method to frequency dynamic analysis. [190] proposed a combined estimation-correction deep learning framework for frequency stability evaluation with a deep neural network (DNN) based estimator to achieve automatic feature extraction and stacked ELM (SELM)-based corrector to reduce the prediction error, and the methods were verified to have great accuracy and superior generalization performance. The DL model, which combines a convolutional neural network (CNN) and a long short-term memory (LSTM) network, has been proposed in [191] to fully exploit both spatial and temporal dynamic measurements, and transfer learning was used to improve the performance of pre-trained DL model in new tasks in this paper.

In general, data-driven intelligent methods have shown great potential in meeting the speed and accuracy requirements of frequency analysis in actual power systems. However, the inherent defects of machine learning, such as poor interpretability, poor generalization, and dependence on sample set data, may lead to purely data-driven methods being unsuitable for RES-dominated power systems with variable operating modes. Conclusions derived from data-driven intelligent methods may only be applicable to specific systems under certain parameters, making it difficult to ensure their accuracy in arbitrary scenarios. This limitation is critical for power systems with strict requirements for safety and reliability. All artificial intelligence methods must be subjected to scrutiny before any real application is attempted. One solution to this problem could be the model-driven and data-driven integrated methods proposed in [192], where the SFR model (belonging to EFM methods) maintains basic physical causality, and the ELM learning method (belonging to intelligent methods) is used to fit and correct errors of the SFR.

The merits and demerits of the whole aforementioned frequency dynamic analysis methods and the application of measurements in those are summarized in Table 3.

5. Conclusion and future work

As the revolution of generation, transmission, demand, and storage side is led by the high integration of RES, the frequency issues of power systems present much more complex features. Therefore, the frequency dynamics monitor and analysis face increasingly severe challenges due to high-dimensional time-varying nonlinear power systems with multi-source interaction. Thereby wide area measurements are introduced to realize the frequency dynamic online monitoring and improve the frequency analysis methods. Firstly, measurements provide valuable information for describing the whole life of frequency dynamics, including the pre-event frequency support (based on inertia estimation technology), the occurrence of frequency deviation (based on frequency event detection technology), the development of frequency dynamics (based on frequency response indicators estimation deviation), and the propagation of frequency wave in wide area system (based on frequency spatio-temporal characteristics monitoring technology). It is evident that the complex frequency dynamic behaviors of RES-dominated power systems can be monitored with abundant measurements in wide area systems. Further, measurements improve the accuracy and reliability of model-driven system frequency security assessment methods by enabling online validation or calibration of dynamic models through parameter identification. They also promote and development of more effective data-driven methods by providing datasets for training, validation, and testing. The application of measurements provides a possible solution for frequency security assessment in the RES-dominated power system with multi-source interaction.

While measurements-based frequency dynamics monitoring and analysis have shown promising research results, there is still much to be explored in the application of measurements to frequency issues. In the following, we will delve into the frequency-related issues that require further investigation in the power system and the technical solutions provided by our research team in this regard.

5.1. *Frequency-related issues to be further investigsted*

- **Frequency dynamics analysis methods with accuracy, speed, and interpretability**

In RES-dominated power system, the new features of frequency dynamics put forward higher requirements for frequency analysis methods: the sharp change of frequency after disturbance and the corresponding serious effects require high accuracy in the analysis methods, the multi-source participation in frequency regulation from

Table 3
Summary of frequency dynamic analysis methods.

Methods		Merits	Demerits	Application of measurements
Model-driven methods	TDS methods	High accuracy	A large amount of computation;Difficult modeling	Validating and calibrating dynamic simulation models
	EFM methods	Clear mechanism; Little computational costs;Easy expansion.	Low solution accuracy;Difficulty in obtaining equivalent parameters	Obtaining model equivalent parameters by system identification/ parameters estimation
Data-driven methods	Intelligent methods	High accuracy and fast speed	Poor interpretability;Poor generalization;	Providing train, validation, and test datasets

generation, transmission, demand, and storage side require extensibility of the analysis methods, and the massive operation scenarios brought by the high uncertainty of renewable energy need fast methods. However, existing time-domain simulation (TDS) methods, equivalent frequency model (EFM) methods, and intelligent methods cannot meet all of these requirements simultaneously. To address this challenge, model-driven and data-driven integrated methods have been proposed as a possible solution [192].

- **Frequency spatial characteristic analysis methods with improved accuracy and calculation speed**

The dispersion of frequency spatio-temporal characteristics is an increasingly prominent feature of system frequency dynamics. However, the awareness of wide area system frequency spatial and temporal distribution mainly relied on the measurement data of frequency or frequency curves obtained from time domain simulations. The present research on evaluating and modeling the spatial distribution of frequency dynamics does not consider the new components of the RES-dominated systems [139–141], as the current means are limited by the system's complexity. Therefore, it is necessary to explore the mechanism of frequency spatial distribution based on measurements and construct an analytical model with improved accuracy and calculation speed to realize online evaluation of the frequency behavior in wide area systems.

- **Parameter identification of RES frequency control**

While studies on parameter identification of RES based on measurements are emerging, the frequency control parameter estimation has not yet been addressed. Regarding the frequency control of RES, the control power is uncontrollable and stochastic, and the strategies and parameters of the controllers can be designed arbitrarily [14]. Thus, the modeling and parameter identification of RES frequency control are challenging tasks, but as more and more RESs and PECs are designed to participate in the frequency response and play a vital role in system frequency dynamics, they will become increasingly important.

- **Interaction analysis of frequency dynamics and system regulation parameters**

Existing research of frequency dynamics monitoring mainly focuses on system frequency dynamics features or frequency response performance respectively. However, the quantification of the interaction between frequency regulation parameters (such as inertia constant, primary frequency regulation time constants, and regulation coefficients) and frequency dynamic behaviors has not been thoroughly explored [193]. It is essential to explore the influence of different frequency regulation solutions, especially from RESs and PECs, on the optimal coordinated control of heterogeneous multiple frequency regulation resources [194]. Given the multiple effects of RES-dominated power systems on frequency dynamic behaviors, it is necessary to rely on mass data based on wide area measurements to mine the relationships between multiple sources.

- **Frequency dynamics analysis considering voltage dynamics impact during severe failure**

The voltage dynamics have been overlooked in the power system dynamics. The power balance in the power system consists of coincidence satisfaction of both active and reactive power equations, not just one. Overlooking the reactive power equations and voltage dynamics is an undeniable challenge for the frequency analysis method in practice. It is commonly recognized in previous studies that voltage dynamics are a key and distinct part of power system stability and control, which is almost independent of frequency dynamics. The new techniques should be investigated and validated through applying them to the proper and accurate enough model of power system considering both voltage and frequency dynamics as key state variables [195,196].

5.2. *Possible solutions in our opinion*

Our team has undertaken preliminary explorations on the topic of measurement-based frequency dynamics modeling and analysis of power systems, and has developed preliminary research results to address the above mentioned issues. We now provide a brief discussion of the potential solution we believe could address these pertinent issues:

(1) For frequency dynamics analysis methods meeting the requirements of RES-dominated power systems, the technical route we constructed is: based on the identification of source-load parameters, we establish a bottom-up framework, encompassing data measurement, parameter identification, equivalent modelling, model aggregation, and real-time application [197]. Source-load frequency-related parameter identification based on measurements improves the accuracy of frequency dynamics analysis by updating the equivalent frequency model parameters online in real time [198]. The aggregation method significantly reduces the complexity of the multi-frequency response resource analysis issues. Our solution facilitates the creation of a more comprehensive and efficient analysis model that accounts for the source-load frequency dynamic behaviors in RES-dominated power system.
(2) For the frequency spatio-temporal dynamic characteristic analysis of wide-area system, we conduct model of the multiple region frequency spatio-temporal dynamics with proposing the method of separating regional system with the same frequency dynamics, and conducting frequency equivalence model within the region and AC/DC interconnection power coupled models between regions. By disregarding the frequency spatial distribution characteristics within co-frequency region and focusing solely on the differences in frequency dynamics among a few regions, this approach significantly reduces the computational dimensions involved in frequency spatiotemporal analysis. Measurements can facilitate the partition of co-frequency regional power system [199]. This approach enables the simultaneous consideration of accuracy and efficiency in the monitoring of frequency spatio-temporal characteristics in wide-area systems.
(3) To address the challenge of constructing accurate physical models for the complex RES system, our envisioned technical

approach involves directly constructing equivalent frequency models of power electronic devices based on measurement data from the the perspective of system frequency dynamics. The relevant parameters of these equivalent models can be identified subsequently with measurements. This approach will provide an intuitive representation of how the dynamic behavior of PEC devices influences the system frequency dynamics and enables real-time tracking of PEC characteristics that are both arbitrary and variable, based on the available data.

(4) Based on the previously conceived accurate and effective frequency analysis model taking into account the spatio-temporal characteristics, we can consider to further constructing the mapping relationship between the regulation parameters and the system frequency dynamics. During the variation of system regulation parameters, we can observe the changes in the identified model parameters and subsequently analyze the alterations in the system frequency behavior.

(5) In addressing the stability analysis challenge that takes into account both voltage and frequency state variables, our future plans involve researching the coupling mechanisms and developmental processes of multi-stability issues under severe faults. We aim to analyze the spatial distribution and temporal characteristics of the dominant processes in the development of various intertwined voltage-frequency instability scenarios. Multivariate measurement data can provide a foundational basis for the analysis of this intricate problem.

CRediT authorship contribution statement

Xuemei Chen: Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing. **Yang Jiang:** Investigation, Methodology, Writing – original draft. **Vladimir Terzija:** Supervision, Resources, Formal analysis, Writing – review & editing. **Chao Lu:** Conceptualization, Funding acquisition, Project administration, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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